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Experiment 06

Aim: To study and Implement Agile Software Project lifecycle using Jira with DevOps Pipeline Integration.

This experiment maps to LO1

LO1: Apply Agile methodologies and use Jira for Sprint Planning, task tracking and integrating with DevOps workflow.

Theory:

What is Jira?

Jira is an Agile project management and issue tracking tool developed by Atlassian. It helps software engineering teams plan sprints, track tasks, manage bugs, and monitor releases across customizable workflows.

Core Concepts & Hierarchy

- Project: A shared workspace representing a product, application, or team initiative.
- Issue Hierarchy:
  - Epic: A large objective or high-level feature broken down into smaller tasks (e.g., *Core Infrastructure*).
  - Story: A end-user feature written from a user perspective (e.g., *Storefront UI*).
  - Task / Bug: Technical implementations or defects to resolve.
  - Sub-task: Granular steps within a story or task.
- Product Backlog: A prioritized list of all pending features, enhancements, and bug fixes awaiting assignment.
- Sprint & Scrum Board: A time-boxed cycle (typically 1–2 weeks) where team members pull items from the backlog into active workflow columns (*To Do*, *In Progress*, *In Review*, *Done*).
- Story Points: Relative estimates assessing effort, technical complexity, and risk for each work item.

Jira Setup & Deployment Models

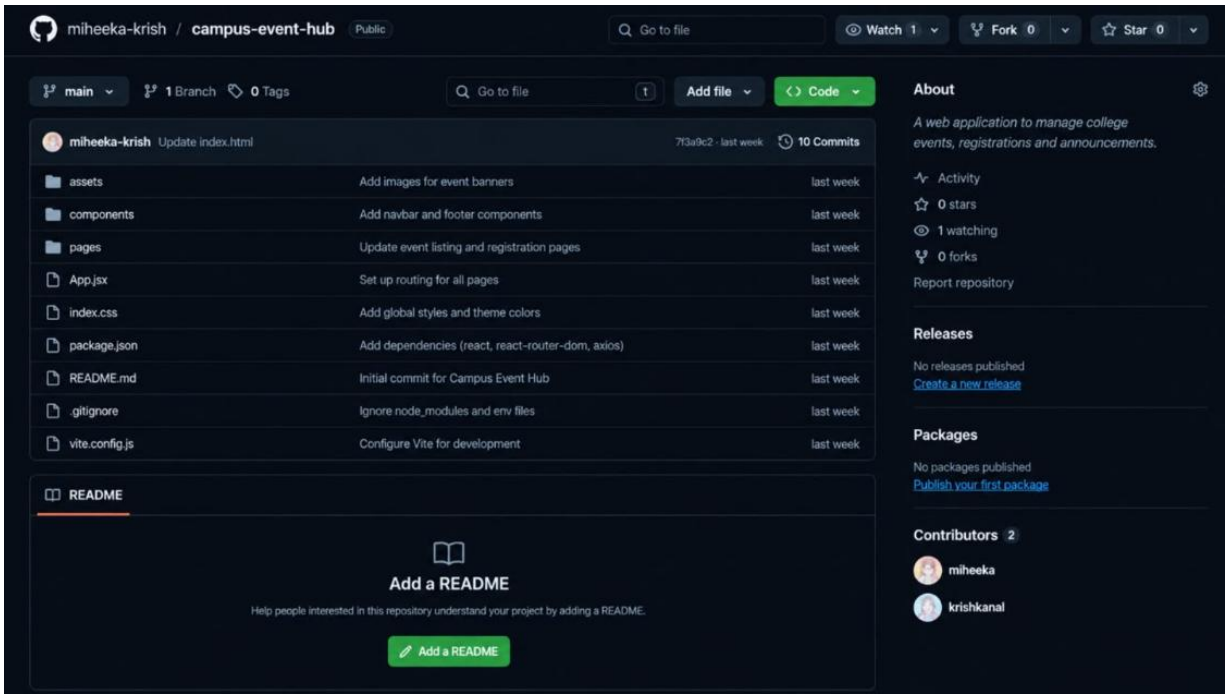
- Jira Cloud (SaaS): Hosted directly by Atlassian. No manual server maintenance or database installations required; teams create an organization site (e.g., company.atlassian.net) and get instant access via a web browser.
- Jira Data Center (Self-Managed): Installed on private infrastructure or dedicated cloud servers (AWS/Azure) for enterprises requiring localized data governance, custom security policies, and high-availability clustering.

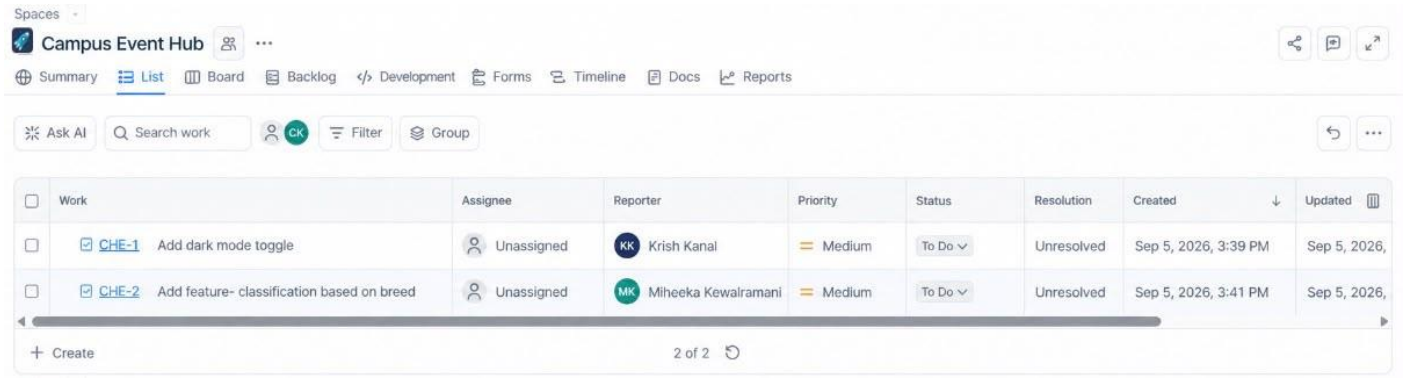
DevOps Integration & Automation Lifecycle

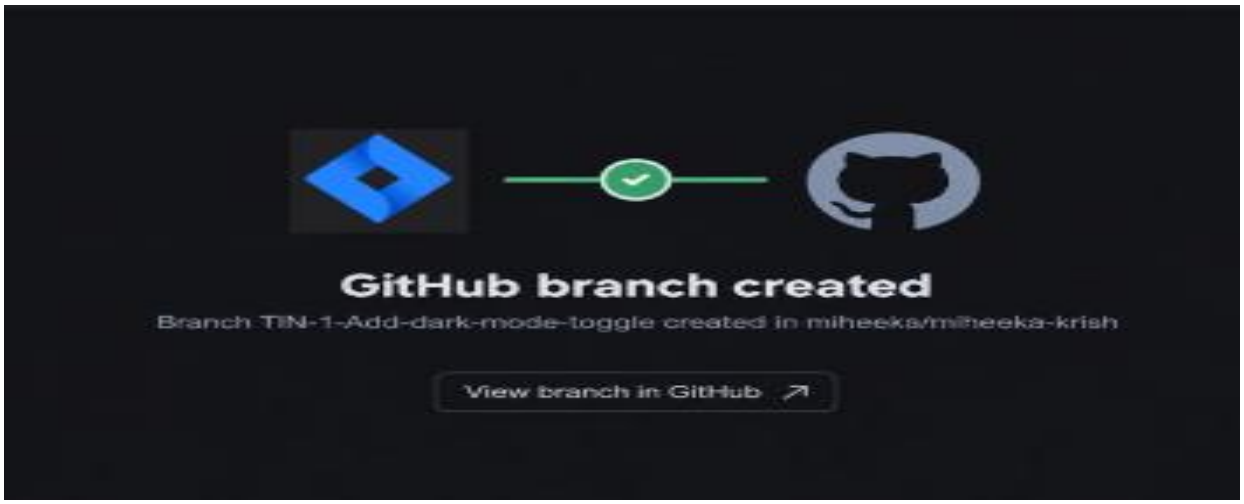
- Version Control Linkage: Integrates with code repositories (such as GitHub, GitLab, or Bitbucket) through official marketplace apps (e.g., *GitHub for Jira*).
- Smart Issue Keys: Mentioning ticket identifiers (e.g., SCRUM-4, SCRUM-6) in branch names, commit messages, or Pull Request titles binds VCS activity directly to the Jira issue panel.
- Jira Automation Rules: Event-driven rules that trigger state transitions across the board (e.g., automatically shifting a ticket to *In Review* when a Pull Request is opened, or transitioning to *Done* upon a merged PR).

Team Collaboration & Access Management

- Role-Based Access Control (RBAC): Access is governed at the organization and project levels via roles (*Administrator*, *Member*, *Viewer*).
- Work Allocation: Tickets are assigned directly to individual developers, allowing visibility into sprint capacity, bottlenecks, and velocity reports.


miheeka-krish / campus-event-hub
Public
Go to file
Watch 1
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
miheeka-krish Update index.html
7f3a9c2 · last week
10 Commits
assets Add images for event banners last week
components Add navbar and footer components last week
pages Update event listing and registration pages last week
App.jsx Set up routing for all pages last week
index.css Add global styles and theme colors last week
package.json Add dependencies (react, react-router-dom, axios) last week
README.md Initial commit for Campus Event Hub last week
.gitignore Ignore node_modules and env files last week
vite.config.js Configure Vite for development last week
README
Add a README
Help people interested in this repository understand your project by adding a README.
Add a README
About
A web application to manage college events, registrations and announcements.
Activity
0 stars
1 watching
0 forks
Report repository
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 2
miheeka
krishkanal


Spaces
Campus Event Hub
Summary
List
Board
Backlog
Development
Forms
Timeline
Docs
Reports
Ask AI
Search work
Filter
Group
Work
Assignee
Reporter
Priority
Status
Resolution
Created
Updated
CHE-1 Add dark mode toggle
Unassigned
Krish Kanal
Medium
To Do
Unresolved
Sep 5, 2026, 3:39 PM
Sep 5, 2026,
CHE-2 Add feature- classification based on breed
Unassigned
Miheeka Kewalramani
Medium
To Do
Unresolved
Sep 5, 2026, 3:41 PM
Sep 5, 2026,
Create
2 of 2


GitHub branch created
Branch TIN-1-Add-dark-mode-toggle created in miheeka/miheeka-krish
View branch in GitHub

Conclusion:

Integrating Jira with GitHub helped us establish an automated and efficient Agile development lifecycle for our E-commerce application. We learned how to manage backlogs, plan active sprints, assign tasks to team members, and monitor progress using Scrum boards. By including Jira issue keys in branch names and Pull Requests, we achieved better traceability between development work and Jira tickets. Additionally, GitHub Actions CI pipelines and Jira Automation helped reduce manual effort by automating testing and updating Jira issues when code was merged. Overall, the integration improved collaboration, tracking, and efficiency throughout the development process.